a

Prediction horizons

True-positive rate

False-positive rate

Random guess
Prediction horizon 12 months
AUROC 0.844
Prediction horizon 18 months
AUROC 0.891
Prediction horizon 24 months
AUROC 0.883

Relative risk

Top % of patients at risk

Prediction horizon 12 months
Prediction horizon 18 months
Prediction horizon 24 months

b

Model types

True-positive rate

False-positive rate

Random guess
Pretrain + Transformer
AUROC 0.891
Pretrain + TimeLSTM
AUROC 0.872
GRU-D:AUROC 0.816
Transformer:AUROC 0.788
XGBoost:AUROC 0.815

Relative risk

Top % of patients at risk

Pretrain + Transformer
Pretrain + TimeLSTM
GRU-D
Transformer
XGBoost

c

Feature tiers

True-positive rate

False-positive rate

Random guess
Standard package
AUROC 0.891
Standard package +Physician selected:AUROC 0.869
Standard package +Extended package:AUROC 0.906

Relative risk

Top % of patients at risk

Standard package
Standard package + Physician-selected
Standard package + Extended package

d

Minimum sequence length

True-positive rate

False-positive rate

Random guess
Sequence length ≥3
AUROC 0.852
Sequence length ≥4
AUROC 0.891
Sequence length ≥5
AUROC 0.886

Relative risk

Top % of patients at risk

Sequence length ≥3
Sequence length ≥4
Sequence length ≥5

e

Missingness tolerance

True-positive rate

False-positive rate

Random guess
Missing rate 0%
AUROC 0.804
Missing rate ≤10%
AUROC 0.857
Missing rate ≤20%
AUROC 0.891
Missing rate ≤30%
AUROC 0.853

Relative risk

Top % patients at risk

Missing rate 0%
Missing rate ≤10%
Missing rate ≤20%
Missing rate ≤30%